

Example PS Response

Applied Stats/Quant Methods 1

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Instructions

This is a template from which you can reference to create your own responses in R and LaTeX.

Question 1

Maybe you need to load in my dataset first, so let's show you how to 'present' that information. You can merely 'show' us that you read in the data by 'printing' your code.

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

Notice, I'm reading in only one line of code from the answers in my .R file using this code:

```
\lstinputlisting[language=R, firstline=40, lastline=40]{PS01_answersJZ.R}}
```

If you're looking at the code in the .tex file as you investigate this example, **which you should**, you'll notice that you could also copy your code results using the `verbatim` environment (`\begin{verbatim}` PASTE RESULTS HERE `\end{verbatim}`). The results will look something like this:

STATE		Y		X1		X2		X3	
AK	: 1	Min.	: 49.00	Min.	:1053	Min.	:334.0	Min.	:326.0
AL	: 1	1st Qu.	: 68.25	1st Qu.	:1698	1st Qu.	:374.2	1st Qu.	:426.2
AR	: 1	Median	: 81.00	Median	:1897	Median	:395.0	Median	:568.0
AZ	: 1	Mean	: 85.04	Mean	:1912	Mean	:404.7	Mean	:561.7
CA	: 1	3rd Qu.	:102.00	3rd Qu.	:2096	3rd Qu.	:419.5	3rd Qu.	:661.2
CO	: 1	Max.	:142.00	Max.	:2817	Max.	:637.0	Max.	:899.0

You can also save figures in R, and place them in your answers that you're writing in your .tex file. First, you need to make sure your path/file name is correct, then you'll save your work when you're in R (see code below).

```

1 # create scatterplot of Y and X1
2 pdf("plot_example.pdf")
3 plot(expenditure$X1, expenditure$Y)
4 dev.off()

```

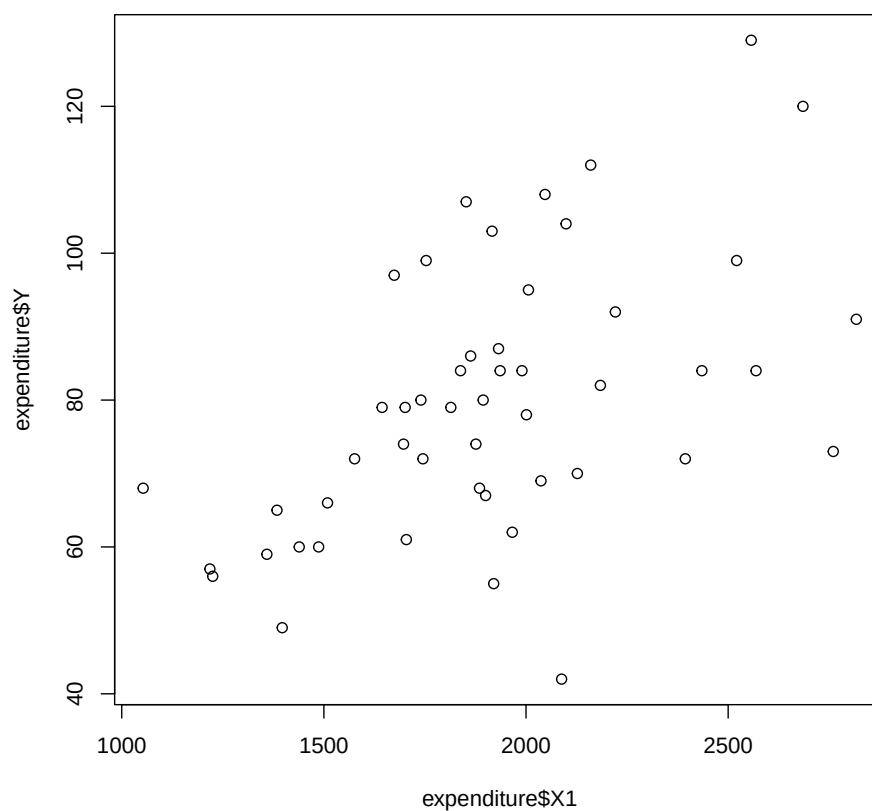
With our figure saved, we just need to render it in our .tex file, which we can do using the `figure` environment:

```

\begin{figure}[h!]\centering
\caption{\footnotesize Example from base plot in R.}
\label{fig:plot_1}
\includegraphics[width=.85\textwidth]{plot_example.pdf}
\end{figure}

```

Figure 1: Example from base plot in R.



Finally can also save tables in R, and place them in your answers in your .tex file, just like you would a figure. You will essentially dump and save the information in a new file, and then read that file in through Latex.

```

1 # run an example regression , to show how to save table
2 regression1 <- lm(Y~X1, data=expenditure)
3 # now save that output to a file that you can read in later to your answers
4 # make it easier for when we need to do this again, let's create a function
5 output_stargazer <- function(outputFile, ...) {
6   output <- capture.output(stargazer(...))
7   cat(paste(output, collapse = "\n"), "\n", file=outputFile, append=TRUE)
8 }
9 # execute function and check ls() to make sure it worked
10 output_stargazer("regression_output1.tex", regression1)

```

That's great, you saved your table in a new file in the same folder as your .tex and .R files. Now, let's read in our saved table using `\inputregression_output1.tex`, which will result in:

Table 1

<i>Dependent variable:</i>	
	Y
X1	0.025*** (0.006)
Constant	32.546*** (11.034)
Observations	50
R ²	0.283
Adjusted R ²	0.268
Residual Std. Error	15.836 (df = 48)
F Statistic	18.920*** (df = 1; 48)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

Or, you can paste the code you get from `stargazer` in R into the `verbatim` environment in LaTeX. This is more labor intensive, but produces the same results.